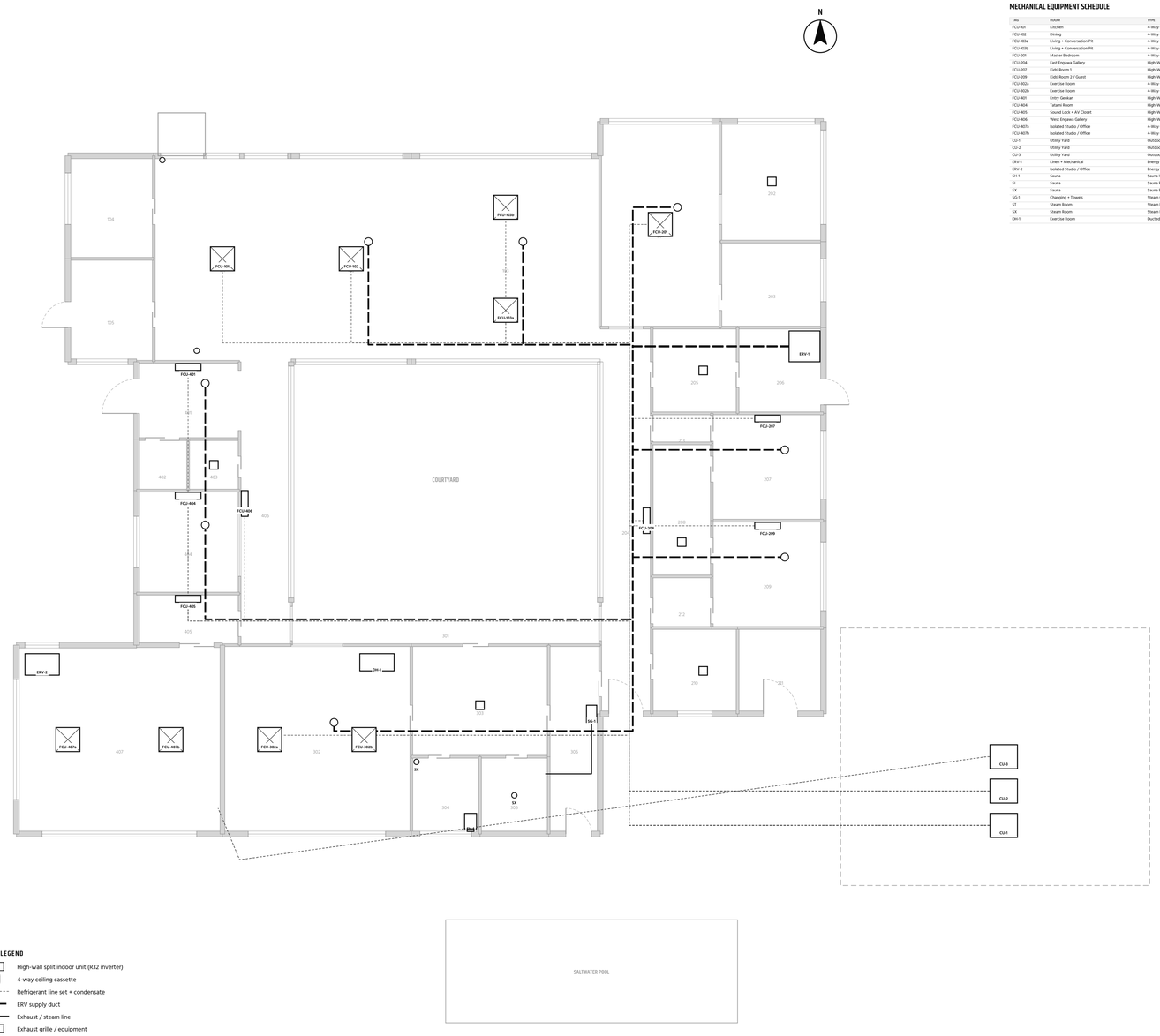

MECHANICAL EQUIPMENT SCHEDULE

TAG	ROOM	TYPE	CAPACITY / SPEC	SYSTEM
FCU-101	Kitchen	4-Way Ceiling Cassette (Inverter, R32)	36,000 BTU/h (10.6 kW)	Multi-Split / VRF CU-1
FCU-102	Dining	4-Way Ceiling Cassette (Inverter, R32)	30,000 BTU/h (8.8 kW)	Multi-Split / VRF CU-1
FCU-102a	Living + Conversation Pit	4-Way Ceiling Cassette (Inverter, R32)	24,000 BTU/h (7.0 kW)	Multi-Split / VRF CU-1
FCU-102b	Living + Conversation Pit	4-Way Ceiling Cassette (Inverter, R32)	24,000 BTU/h (7.0 kW)	Multi-Split / VRF CU-1
FCU-201	Master Bedroom	4-Way Ceiling Cassette (Inverter, R32)	30,000 BTU/h (8.8 kW)	Multi-Split / VRF CU-2
FCU-204	East Engawa Gallery	High-Wall Split Indoor Unit (Inverter, R32)	24,000 BTU/h (7.0 kW)	Multi-Split / VRF CU-2
FCU-207	Kids' Room 1	High-Wall Split Indoor Unit (Inverter, R32)	18,000 BTU/h (5.3 kW)	Multi-Split / VRF CU-2
FCU-209	Kids' Room 2 / Guest	High-Wall Split Indoor Unit (Inverter, R32)	18,000 BTU/h (5.3 kW)	Multi-Split / VRF CU-2
FCU-302a	Exercise Room	4-Way Ceiling Cassette (Inverter, R32)	24,000 BTU/h (7.0 kW)	Multi-Split / VRF CU-2
FCU-302b	Exercise Room	4-Way Ceiling Cassette (Inverter, R32)	24,000 BTU/h (7.0 kW)	Multi-Split / VRF CU-2
FCU-401	Entry Genkan	High-Wall Split Indoor Unit (Inverter, R32)	9,000 BTU/h (2.6 kW)	Multi-Split / VRF CU-1
FCU-404	Tatami Room	High-Wall Split Indoor Unit (Inverter, R32)	12,000 BTU/h (3.5 kW)	Multi-Split / VRF CU-1
FCU-405	Sound Lock + AV Closet	High-Wall Split Indoor Unit (Inverter, R32)	9,000 BTU/h (2.6 kW)	Multi-Split / VRF CU-1
FCU-406	West Engawa Gallery	High-Wall Split Indoor Unit (Inverter, R32)	18,000 BTU/h (5.3 kW)	Multi-Split / VRF CU-1
FCU-407a	Isolated Studio / Office	4-Way Ceiling Cassette (Inverter, R32)	24,000 BTU/h (7.0 kW)	Multi-Split / VRF CU-3
FCU-407b	Isolated Studio / Office	4-Way Ceiling Cassette (Inverter, R32)	24,000 BTU/h (7.0 kW)	Multi-Split / VRF CU-3
CU-1	Utility Yard	Outdoor Heat-Pump Unit CU-1	162,000 BTU/h connected (47.5 kW), inverter VRF/multi-split	Multi-Split / VRF CU-1
CU-2	Utility Yard	Outdoor Heat-Pump Unit CU-2	150,000 BTU/h connected (43.9 kW), inverter VRF/multi-split	Multi-Split / VRF CU-2
CU-3	Utility Yard	Outdoor Heat-Pump Unit CU-3	48,000 BTU/h connected (14.1 kW), inverter VRF/multi-split	Multi-Split / VRF CU-3
ERV-1	Linen + Mechanical	Energy Recovery Ventilator ERV-1	350 m³/h, enthalpy core, MERV 13 + PM2.5 filter	Ventilation (ERV)
ERV-2	Isolated Studio / Office	Energy Recovery Ventilator ERV-2 (Studio)	120 m³/h, acoustic-lined ducts	Ventilation (ERV)
SH-1	Sauna	Sauna Heater SH-1	9 kW electric rock heater, 400 V 3-phase, guard rail	Sauna
SI	Sauna	Sauna Fresh-Air Inlet (Low, Behind Heater)	100 mm, from crawl / exterior	Sauna
SX	Sauna	Sauna Exhaust (High, Opposite Wall) + Fan	100 mm, 2 air changes/h, to outside	Sauna
SG-1	Changing + Towels	Steam Generator SG-1	9 kW, 400 V 3-phase, auto-flush, in Changing Room cabinet	Steam
ST	Steam Room	Steam Line to Steam Head	15 mm copper, sloped to head, insulated	Steam
SX	Steam Room	Steam Room Exhaust Fan (Post-Use Purge)	40 L/s, timer, to outside	Steam
DH-1	Exercise Room	Ducted Dehumidifier DH-1	90 L/day, for wet-room + gym humidity	Dehumidification

HVAC LEGEND

- High-wall split indoor unit (R32 inverter)
- 4-way ceiling cassette
- Refrigerant line set + condensate
- ERV supply duct
- Exhaust / steam line
- Exhaust grille / equipment

1 HVAC PLAN

SCALE: 1/8" = 1'-0"

WEBERMAN RESIDENCE

Mountain courtyard house, Thailand
Owner: Max Weberman

ISSUE

Design Development

NOT FOR CONSTRUCTION

DATE
11 Oct 2026

SCALE
1/8" = 1'-0"

MODEL
bim.json (from plan.json)

UNITS
Feet + inches (mm for members)

REVISION
DD-01

HVAC NOTES

Cooling sized at about 850 BTU/h per m² (Thai rule of thumb for glazed rooms); verify with a heat-load calc.

ERV runs continuously for fresh air + humidity control; bath boost on humidity sensor.

Sauna: low inlet behind heater, high exhaust opposite. Steam: vapor-sealed, sloped ceiling, purge fan.

SHEET TITLE
HVAC Plan

SHEET
M-101